

Hexadecimal to Binary & Binary to Hexadecimal Conversion

$$r = 16$$

>> In hexadecimal number system we have 16 different digits.

>> Each one of them can be represented by a equivalent 4-bit binary number.

$$0 \neq (16-1)$$

$$0 \neq 15$$

0	-	0000
1	-	0001
2	-	0010
3	-	0011
4	-	0100
5	-	0101
6	-	0110
7	-	0111
8	-	1000
9	-	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Hex to Binary :-

i) $(259A)_{16} \rightarrow (0010\ 0101\ 1001\ 1010)_2$

Diagram showing the conversion of each hex digit to its 4-bit binary equivalent:

2 → 0010, 5 → 0101, 9 → 1001, A → 1010

ii) $(CAFE.3D)_{16} \rightarrow (1100\ 1010\ 1111\ 1110.0011\ 1101)_2$

Diagram showing the conversion of each hex digit to its 4-bit binary equivalent:

C → 1100, A → 1010, F → 1111, E → 1110, 3 → 0011, D → 1101



11	B	-	1011
12	<u>C</u>	-	1100
13	D	-	1101
14	E	-	1110
15	F	-	1111

Binary to Hex :-

i) $0001\ 1000\ 1001\ .\ 1100 \rightarrow (189.C)_{16}$

(Note: The original image has some additional markings: a blue '1' above the first '0' of the first group, a pink bracket under the first group, a pink bracket under the second group, a pink bracket under the third group, a pink bracket under the fourth group, and a pink '2' below the fourth group.)

4w i) $(FACE)_{16} \rightarrow (\quad)_2$

ii) $(1101\ 1110\ 1010\ 1111)_2 \rightarrow (\quad)_{16}$

(Note: A mouse cursor is pointing at the first '1' of the second group '1110' in the binary string.)

1. 1111 1010 1100 1110
2. DEAF